Exp -8

```
clc;
clear;
close all;

l = 30;
r = 15;

figure;
grid on;
hold on;
axis equal;
axis([-20 50 -20 50]);

angle = 0:0.01:2*pi;
x_circle = r*cos(angle);
y_circle = r*sin(angle);

plot(x_circle,y_circle,'k','LineWidth',2);

for A = 0:0.08:4*pi

x = r*cos(A) + sqrt(l^2-r^2*sin(A)^2);

X = [0 r*cos(A) x];
Y = [0 r*sin(A) 0];

cla;
plot(x_circle,y_circle,'k','LineWidth',2);
hold on;

plot(X,Y,'b-o','LineWidth',4);
plot(x,0,'rs','MarkerSize',12,...
'MarkerFaceColor','r');

axis equal;
axis([-20 50 -20 50]);
grid on;

title('Slider Crank Mechanism Animation');
xlabel('X Position');
ylabel('Y Position');

drawnow;
pause(0.1);

end
```

Exp-4

```
A = [1/(1000*9.81), 1/(2*9.81);
1/(1000*9.81), 1/(8*9.81)];

B = [15.2905; 11.4679];

X = A\B;

Pressure = X(1,1)
Velocity_squared = X(2,1)
Velocity = sqrt(Velocity_squared)
```

Exp-5

```
clc;
clear;
close all;

% Experimental data
T = [0 5 10 20 30 40];
mu = [1.787 1.519 1.307 1.002 0.7975 0.6539];

% Convert temperature to Kelvin
Ta = T + 273;

% Arrhenius equation
% mu = B * exp(B1/Ta)
% ln(mu) = ln(B) + B1*(1/Ta)

X = 1./Ta;
Y = log(mu);

% Curve fitting using linear regression
P = polyfit(X,Y,1);

B1 = P(1);
lnB = P(2);
B = exp(lnB);

% Predicted viscosity values
mu_fit = B * exp(B1./Ta);

% Display results
fprintf('Value of B = %f\n',B);
fprintf('Value of B1 = %f\n',B1);

% Plot experimental and fitted data
plot(T,mu,'ro','MarkerSize',8,'LineWidth',2);
hold on;

plot(T,mu_fit,'b-','LineWidth',2);

xlabel('Temperature (C)');
ylabel('Viscosity (mu)');
```

```
title('Effect of Temperature on Viscosity');
legend('Experimental Data','Curve Fit');
grid on;
```

Exp-6

```
clc;
clear;
close all;

Q = [10 20 30 40 50 60 70];
P = [15 23 45 61 75 83 105];

coefficients = polyfit(Q,P,1);

m = coefficients(1);
c = coefficients(2);

Q_fit = linspace(min(Q),max(Q),100);
P_fit = polyval(coefficients,Q_fit);

figure;
plot(Q,P,'ro','MarkerSize',8,'MarkerFaceColor','r');
hold on;
plot(Q_fit,P_fit,'b-','LineWidth',2);

title('Linear Regression: Steady Flow in a Circular Pipe');
xlabel('Flow Rate (L/min)');
ylabel('Pressure Drop (kPa)');

legend('Experimental Data','Linear Fit','Location','northwest');

grid on;
hold off;
```

Exp- 8

```
function drag = aerofun(v)
   sigma = 0.6;
   w = 16000;
   drag = (0.01*sigma*v^2) + (0.95*w^2/(sigma*v^2));
end


[velocity, drag] = fminsearch(@aerofun, 8);

fprintf('Optimal velocity = %.4f\n', velocity);
fprintf('Minimum drag = %.4f\n', drag);
```

Exp-7

```
function d2y = odefun(x,y)
d2y = zeros(2,1);
d2y(1) = y(2);
d2y(2) = -25000*(3000-x)/((2.1e5)*1e8);
end
```

```
[x,y] = ode45(@odefun,[0,3000],[0,0]);
plot(x,y(:,1))
```